AIMER-Q: AI-Driven Adaptive Trust Resilience for Quantum-Safe E-Business Systems*

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Abstract

The convergence of 6G, quantum computing, and AI-driven services introduces major security challenges for future E-Business Information Systems (EBIS). This work proposes AIMER-Q, a reinforcement learning–driven mitigation framework for hybrid PQC–QKD networks using the Quantum Attack Surface Index (QASI) and a Deep Q-Network (DQN) agent for adaptive exposure control. Experimental results show >60% exposure reduction and approximately 3× higher survival probability than static and rule-based baselines.

* Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. P-21, July 8–10, 2026, Seoul, South Korea.
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1 Introduction

Hybrid PQC–QKD architectures are promising for quantum-safe EBIS, but cross-layer interactions introduce dynamic exposure risks including entropy instability, entanglement degradation, replay attacks, and timing leakage. This work proposes the first unified framework combining exposure modeling, probabilistic risk interpretation, and RL-driven adaptive mitigation for hybrid quantum-secure networks.

2 Research Gaps & Contributions

Existing studies separately evaluate PQC, QKD, and AI security, leaving three gaps: absence of composite exposure modeling, RL-driven adaptive mitigation, and probabilistic exposure-risk interpretation. This work introduces QASI, hazard-based risk modeling, DQN-driven mitigation, and empirical validation showing >60% exposure reduction with 3× survival improvement.

3 System Model & Threat Formulation

The hybrid network $H(t) = \{L_{pqc}(t), L_{qkd}(t), L_{cls}(t)\}$ consists of PQC, QKD, and classical control layers whose interdependencies create dynamic cross-layer exposure. PQC entropy degradation, QKD instability, and replay/timing attacks collectively increase adversarial risk.

4 Quantum Attack Surface Index (QASI)

QASI model hybrid PQC-QKD exposure using 3 normalized signals: PQC entropy instability $E_1(t)$, QKD entanglement degradation $E_2(t)$, and classical channel vulnerability $E_3(t)$, $Q(t) = \Sigma^l w^l \cdot E^l(t)$, $Q(t) \in [0,1]$. Adversial success probability is quantified thrpugh the hazard model: $\lambda(t) = \lambda_0 \cdot exp(\beta \cdot Q(t))$. which enables QASI to function as a unified adaptive control variable for exposure-aware mitigation.

5 AIMER-Q: RL-Driven Mitigation Framework

AIMER-Q formulates adaptive defense as a Markov Decision Process where actions include PQC refresh, QKD rekeying, and channel filtering to minimize QASI. A Deep Q-Network continuously learns optimal mitigation policies under dynamic attacks.

6 Experimental Results

AIMER-Q achieves >60% cumulative exposure reduction and approximately 3× higher survival probability while maintaining consistently lower exposure than static and rule-based defenses.

7 E-Business impact

AIMER-Q enables adaptive protection for 6G e-commerce, quantum-safe finance, enterprise IoT, and post-quantum regulatory compliance. The framework shifts EBIS security from static protection toward adaptive quantum-safe resilience.

8 Conclusions & Future Works

AIMER-Q demonstrates that RL-driven exposure mitigation significantly improves resilience in hybrid PQC–QKD environments. Future work includes adaptive QASI weighting, real-world 6G validation, and multi-objective RL considering latency and business SLA constraints.

[1] K. Cherkaoui Dekkaki et al. Exploring Post-Quantum Cryptography. Technologies, 12(12):241, 2024.